

Questions

MathonGo

Q1 - 2024 (04 Apr Shift 2)

Given that the inverse trigonometric function assumes principal values only. Let x, y be any two real numbers in $[-1, 1]$ such that $\cos^{-1} x - \sin^{-1} y = \alpha$, $\frac{-\pi}{2} \leq \alpha \leq \pi$.

Then, the minimum value of $x^2 + y^2 + 2xy \sin \alpha$ is

- (1) 0
- (2) -1
- (3) $\frac{1}{2}$
- (4) $-\frac{1}{2}$

Q2 - 2024 (06 Apr Shift 1)

For $n \in \mathbb{N}$, if $\cot^{-1} 3 + \cot^{-1} 4 + \cot^{-1} 5 + \cot^{-1} n = \frac{\pi}{4}$, then n is equal to _____

Q3 - 2024 (09 Apr Shift 2)

The integral $\int_{1/4}^{3/4} \cos\left(2 \cot^{-1} \sqrt{\frac{1-x}{1+x}}\right) dx$ is equal to

- (1) $1/2$
- (2) $-1/2$
- (3) $-1/4$
- (4) $1/4$

Q4 - 2024 (09 Apr Shift 2)

Let the inverse trigonometric functions take principal values. The number of real solutions of the equation

$2 \sin^{-1} x + 3 \cos^{-1} x = \frac{2\pi}{5}$, is _____

Questions

MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

Answer Key

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

Q1 (1) MathonGo Q2 (47) MathonGo Q3 (3) MathonGo Q4 (0) MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

Solutions

MathonGo

Q1

$$\cos^{-1} x - \left(\frac{\pi}{2} - \cos^{-1} y \right) = \alpha$$

$$\cos^{-1} x + \cos^{-1} y = \frac{\pi}{2} + \alpha$$

$$\alpha \in \left[-\frac{\pi}{2}, \pi \right], \frac{\pi}{2} + \alpha \in \left[0, \frac{3\pi}{2} \right]$$

$$\cos^{-1} \left(xy - \sqrt{1-x^2} \sqrt{1-y^2} \right) = \frac{\pi}{2} + \alpha$$

$$xy - \sqrt{1-x^2} \sqrt{1-y^2} = -\sin \alpha$$

$$(xy + \sin \alpha) = (1-x^2)(1-y^2)$$

$$x^2 y^2 + 2xy \sin \alpha + \sin^2 \alpha = 1 - x^2 - y^2 + x^2 y^2$$

$$x^2 + y^2 + 2xy \sin \alpha = 1 - \sin^2 \alpha$$

$$x^2 + y^2 + 2xy \sin \alpha = \cos^2 \alpha$$

$$\text{Min. value of } \cos^2 \alpha = 0$$

$$\text{At } \alpha = \frac{\pi}{2}$$

Q2

$$\cot^{-1} 3 + \cot^{-1} 4 + \cot^{-1} 5 + \cot^{-1} n = \frac{\pi}{4}$$

$$\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{n} = \frac{\pi}{4}$$

$$\tan^{-1} \left(\frac{46}{48} \right) + \tan^{-1} \frac{1}{n} = \frac{\pi}{4}$$

$$\tan^{-1} \left(\frac{23}{24} \right) + \tan^{-1} \frac{1}{n} = \frac{\pi}{4}$$

$$\tan^{-1} \frac{1}{n} = \tan^{-1} 1 - \tan^{-1} \frac{23}{24}$$

$$\tan^{-1} \frac{1}{n} = \tan^{-1} 1 - \tan^{-1} \frac{23}{24}$$

$$\tan^{-1} \frac{1}{n} = \tan^{-1} \left(\frac{1 - \frac{23}{24}}{1 + \frac{23}{24}} \right)$$

$$\tan^{-1} \frac{1}{n} = \tan^{-1} \left(\frac{\frac{1}{24}}{\frac{47}{24}} \right)$$

$$\tan^{-1} \frac{1}{n} = \tan^{-1} \frac{1}{47}$$

$$n = 47$$

Q3

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

$$\begin{aligned}
 I &= \int_{1/4}^{3/4} \cos \left(2 \cot^{-1} \left(\sqrt{\frac{1-x}{1+x}} \right) dx \right) \\
 &= \int_{1/4}^{3/4} \cos \left(2 \left(\tan^{-1} \sqrt{\frac{1+x}{1-x}} \right) \right) dx \\
 &= \int_{1/4}^{3/4} \frac{1 - \tan^2 \left(\tan^{-1} \sqrt{\frac{1+x}{1-x}} \right)}{1 + \tan^2 \left(\tan^{-1} \sqrt{\frac{1+x}{1-x}} \right)} dx \\
 &= \int_{1/4}^{3/4} \frac{1 - \left(\frac{1+x}{1-x} \right)}{1 + \left(\frac{1+x}{1-x} \right)} dx = \int_{1/4}^{3/4} \frac{-2x}{2} dx \\
 &= \int_{1/4}^{3/4} (-x) dx = - \left(\frac{x^2}{2} \right)_{1/4}^{3/4} \\
 &= - \frac{1}{2} \left[\frac{9}{16} - \frac{1}{16} \right] \\
 &= - \frac{1}{4}
 \end{aligned}$$

Q4

$$\begin{aligned}
 2 \sin^{-1} x + 3 \cos^{-1} x &= \frac{2\pi}{5} \\
 \Rightarrow \pi + \cos^{-1} x &= \frac{2\pi}{5} \\
 \Rightarrow \cos^{-1} x &= \frac{-3\pi}{5}
 \end{aligned}$$

Not possible

Ans. 0

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)